

GEOGRAPHY

INDIAN GEOGRAPHY



INDIAN CLIMATE

MODULE - 5

INDIAN MONSOON

- **THE RHYTHM OF SEASONS**

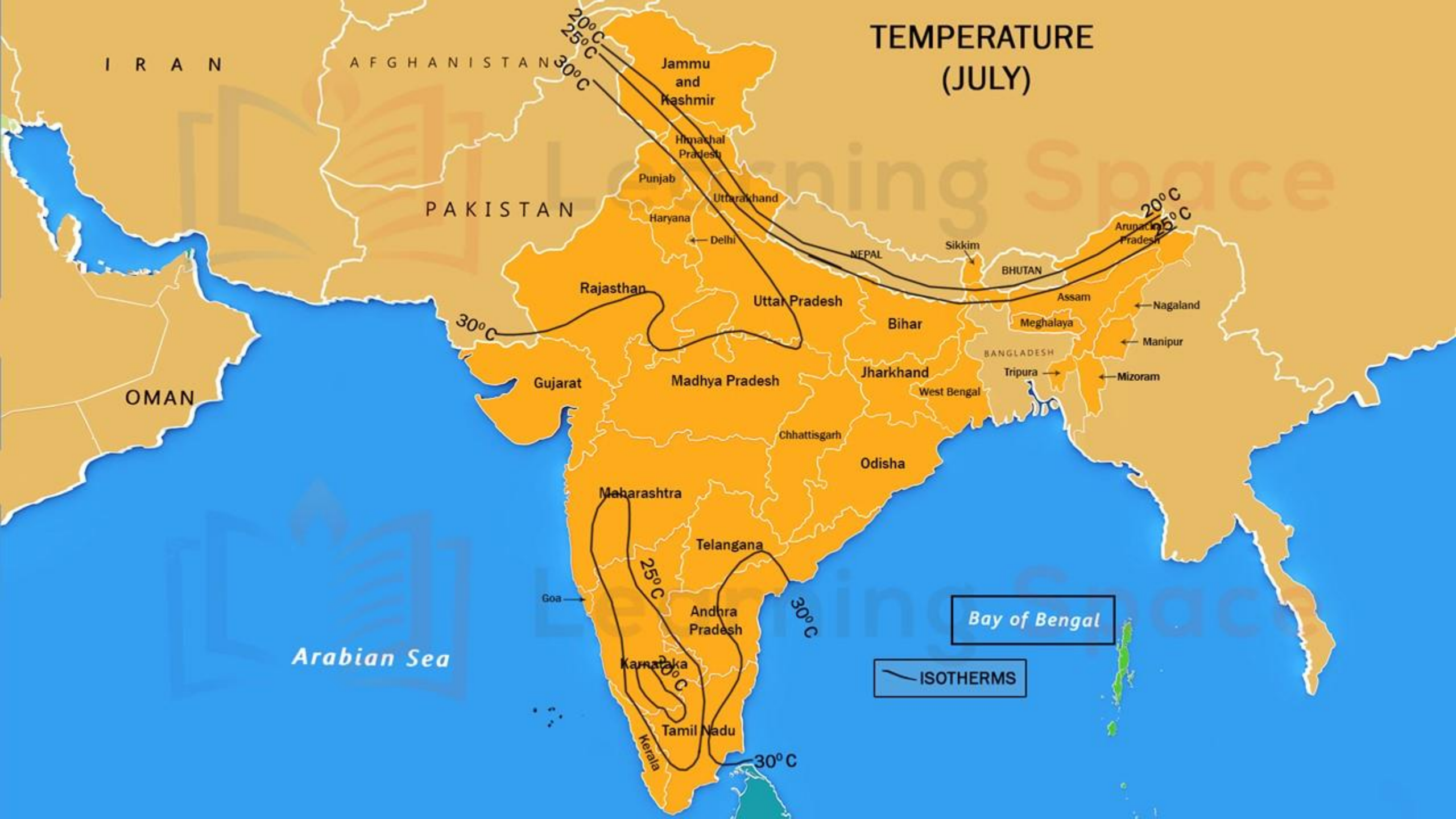
THE RHYTHM OF SEASONS

2

THE HOT WEATHER SEASON

TEMPERATURE

- 1 With the apparent northward movement of the sun towards the Tropic of Cancer in March, temperatures start rising in north India.
- 2 April, May and June are the months of summer in north India. In most parts of India, temperatures recorded are between 30° - 32°C .
- 3 In March, the highest day temperature of about 38°C occurs in the Deccan Plateau while in April, temperature ranging between 38°C and 43°C are found in Gujarat and Madhya Pradesh.
- 4 In May, the heat belt moves further north, and in the north-western part of India, temperatures around 48°C are not uncommon.
- 5 The hot weather season in south India is mild and not so intense as found in north India.
- 6 The Peninsular situation of south India with moderating effect of the oceans keeps the temperatures lower than that prevailing in north India. So, temperatures remain between 26°C and 32°C .



2

THE RHYTHM OF SEASONS

THE HOT WEATHER SEASON

TEMPERATURE

Why monsoon occurs only in the month of June? Why not in April or May?



- In May, the heat belt moves further north, and in the north-western part of India, temperatures around 48°C are not uncommon
- The hot weather season in south India is mild and not so intense as found in north India.
- The southern parts of the country are distinctly warmer in March and April whereas in June, north India has higher temperatures.

The highest temperatures are recorded just before the onset of the southwest monsoons in parts of Northern India in late May which will lead to onset of monsoons.

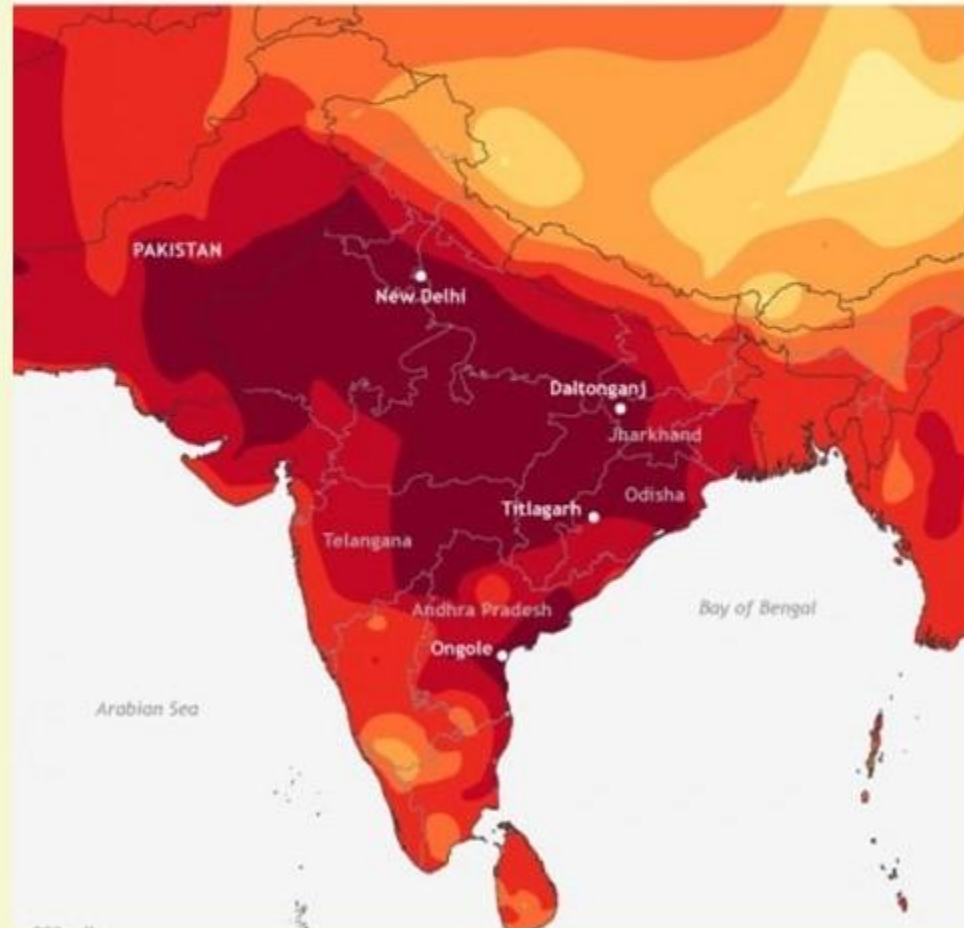
2

THE RHYTHM OF SEASONS

THE HOT WEATHER SEASON

TEMPERATURE

Heat wave (May 24–30, 2015)



7

Due to altitude, the temperatures in the hills of Western Ghats remain below 25°C.

8

In the coastal regions, the north-south extent of isotherms parallel to the coast confirms that temperature does not decrease from north to south rather it increases from the coast to the interior.

9

The mean daily minimum temperature during the summer months also remains quite high and rarely goes below 26°C.

2

THE RHYTHM OF SEASONS

THE HOT WEATHER SEASON

PRESSURE AND WINDS



1 The summer months are a period of excessive heat and falling air pressure in the northern half of the country.

2 Because of the heating of the subcontinent, the **ITCZ** moves northwards occupying a position centred at 25°N in July.

3 Roughly, this elongated low pressure monsoon trough extends over the Thar desert in the north-west to Patna and Chotanagpur in the east-southeast.

PRESSURE AND SURFACE WINDS (JULY)

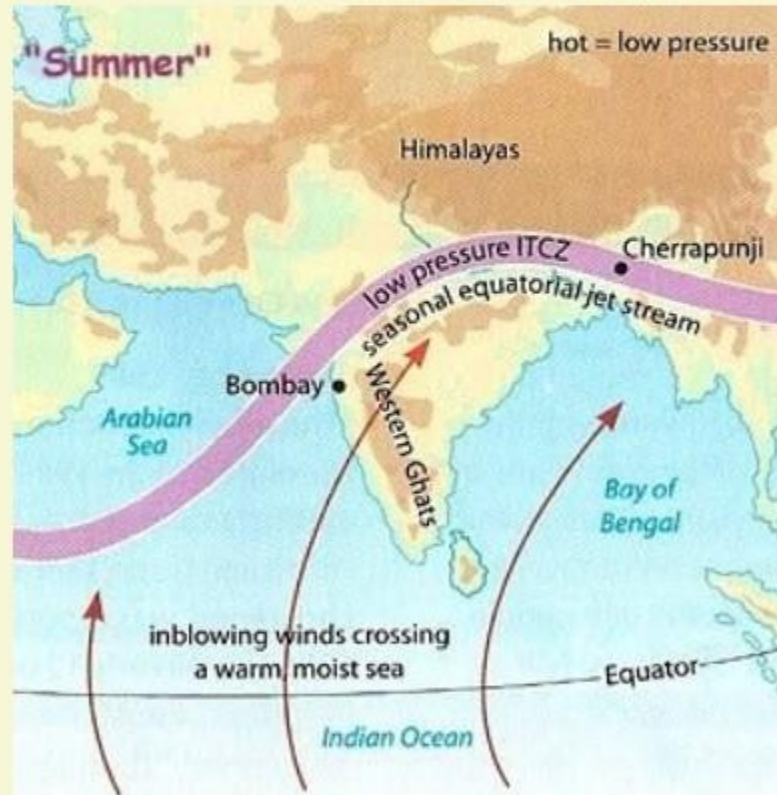


2

THE RHYTHM OF SEASONS

THE HOT WEATHER SEASON

PRESSURE AND WINDS



4 The location of the ITCZ attracts a surface circulation of the winds which are southwesterly on the west coast as well as along the coast of West Bengal and Bangladesh.

5 They are easterly or southeasterly over north Bengal and Bihar.

6 The influx of these winds by mid-June brings about a change in the weather towards the rainy season.

THE RHYTHM OF SEASONS

2

THE HOT WEATHER SEASON

PRESSURE AND WINDS

- 7** In the heart of the ITCZ in the northwest, the dry and hot winds known as 'Loo', blow in the afternoon, and very often, they continue well into midnight.
- 8** Dust storms in the evening are very common during May in Punjab, Haryana, Eastern Rajasthan and Uttar Pradesh.
- 9** These temporary storms bring a welcome respite from the oppressing heat since they bring with them light rains and a pleasant cool breeze.
- 10** Occasionally, the moisture-laden winds are attracted towards the periphery of the trough.
- 11** A sudden contact between dry and moist air masses gives rise to local storms of great intensity.
- 12** These local storms are associated with violent winds, torrential rains and even hailstorms.

Further reference:

<https://india.mongabay.com/2018/05/08/dust-storms-may-increase-in-india-due-to-climate-change/>

PRESSURE AND SURFACE WINDS (JULY)



2

THE RHYTHM OF SEASONS

THE HOT WEATHER SEASON

SOME FAMOUS LOCAL STORMS OF HOT WEATHER SEASON



MANGO SHOWER

- Towards the end of summer, there are pre-monsoon showers which are a common phenomena in Kerala and coastal areas of Karnataka.
- Locally, they are known as mango showers since they help in the early ripening of mangoes.

BLOSSOM SHOWER

With this shower, coffee flowers blossom in Kerala and nearby areas.

2

THE RHYTHM OF SEASONS

THE HOT WEATHER SEASON

SOME FAMOUS LOCAL STORMS OF HOT WEATHER SEASON

NOR WESTERS

- These are **dreaded evening thunderstorms** in Bengal and Assam.
- Their notorious nature can be understood from the local nomenclature of '**Kalbaisakhi**', a calamity of the month of Baisakh.
- These showers are useful for tea, jute and rice cultivation. In Assam, these storms are known as "**Bardoli Chheerha**".

LOO

- Hot, dry and oppressing winds blowing in the Northern plains from Punjab to Bihar with higher intensity between Delhi and Patna.



THE RHYTHM OF SEASONS

3

SOUTHWEST MONSOON SEASON



1

As a result of rapid increase of temperature in May over the north-western plains, the low pressure conditions over there get further intensified.

2

By early June, they are powerful enough to attract the trade winds of Southern Hemisphere coming from the Indian Ocean.

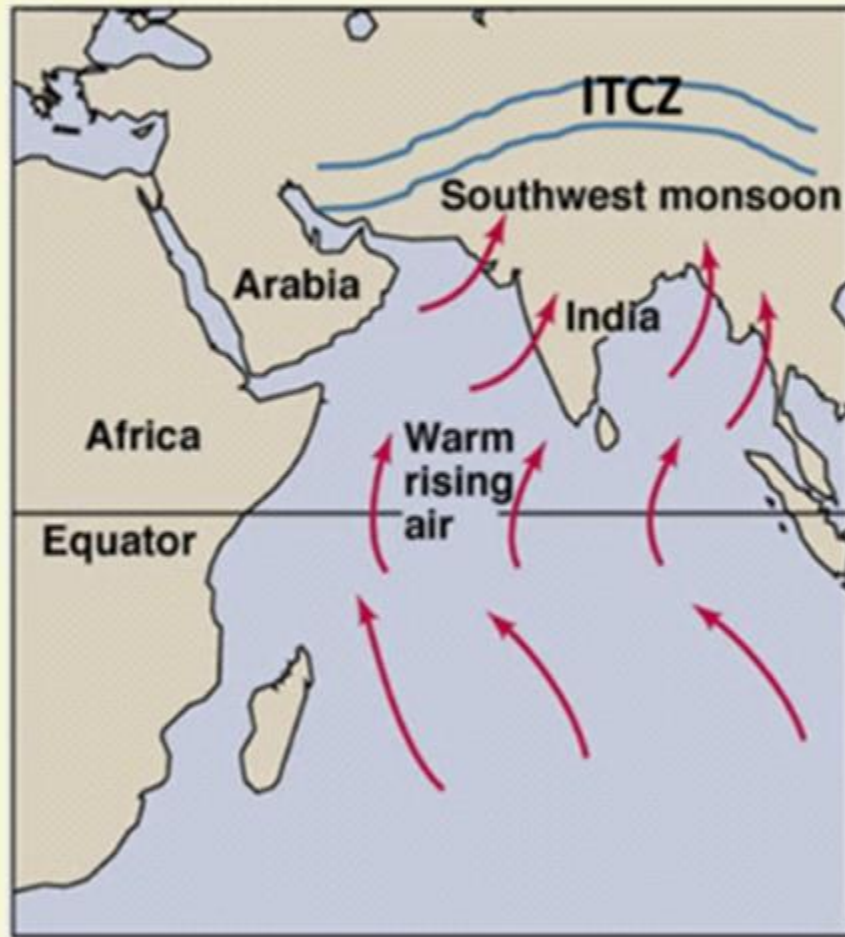
3

These southeast trade winds cross the equator and enter the Bay of Bengal and the Arabian Sea, only to be caught up in the air circulation over India.

THE RHYTHM OF SEASONS

3

SOUTHWEST MONSOON SEASON



SOUTH-WEST MONSOON

4

Passing over the equatorial warm currents, they bring with them moisture in abundance.

5

After crossing the equator, they follow a southwesterly direction. That is why they are known as southwest monsoons.

6

The rain in the southwest monsoon season begins rather abruptly.

7

One result of the first rain is that it brings down the temperature substantially.

THE RHYTHM OF SEASONS

3

SOUTHWEST MONSOON SEASON



8

This sudden onset of the moisture-laden winds associated with violent thunder and lightening, is often termed as the “burst” of the monsoons.

9

The monsoon may burst in the first week of June in the coastal areas of Kerala, Karnataka, Goa and Maharashtra while in the interior parts of the country, it may be delayed to the first week of July.

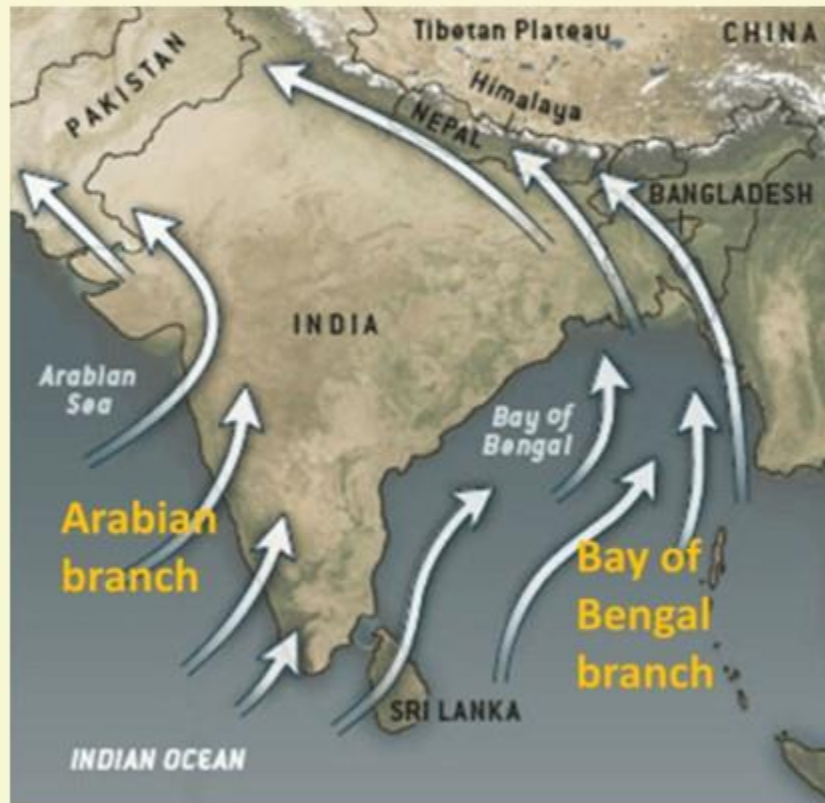
10

The day temperature registers a decline of 5°C to 8°C between mid-June and mid-July.

THE RHYTHM OF SEASONS

3

SOUTHWEST MONSOON SEASON



11

As these winds approach the land, their southwesterly direction is modified by the relief and thermal low pressure over the northwest India.

12

The monsoon approaches the landmass in two branches:
(i) The Arabian Sea branch
(ii) The Bay of Bengal branch.

WE EXPRESS OUR SINCERE THANKS FOR VIEWING THIS VIDEO

Presented by



Learning Space

For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

OUR TEAM

G. V. Rao
K. Srikanth
D. Sunil Kumar
L. V. Krishna
Y. Deepthi

Amrita Naidu
M. Binukrishna
G. Ravi Babu
D. Joji
K. Victor Babu

Visit us at:

www.learningspacedigital.com

0 8 6 6 -2 4 4 4 4 7 2
0 9 8 4 9 9 4 2 2 9 9